



Animate with Scratch

03

The school's Space Club is celebrating "National Astronaut Day" this week. I want to tell the story of Neil Armstrong, the first man on moon.

How do I make an animation?

Adil, why don't you make an animation to tell the story?



Animation is a way to add movement to drawings, models, and non-living objects. This makes them appear to move as if they are alive. Animations are important because we can show our imagination or stories to anyone and they can understand it better.

We can use a software like Scratch to create an animation.

3. Animate with Scratch

We will explore the various parts of Scratch.

- **Scratch:** Scratch is a software that uses blocks in a particular order to give instructions. The instructions can be used to create an animation, story, or game. Sprites and Codes are used as operating tools in Scratch.

- **Sprite:** A sprite can be an animated character, a player in a game, or any other object.

The default sprite in Scratch is a cat.

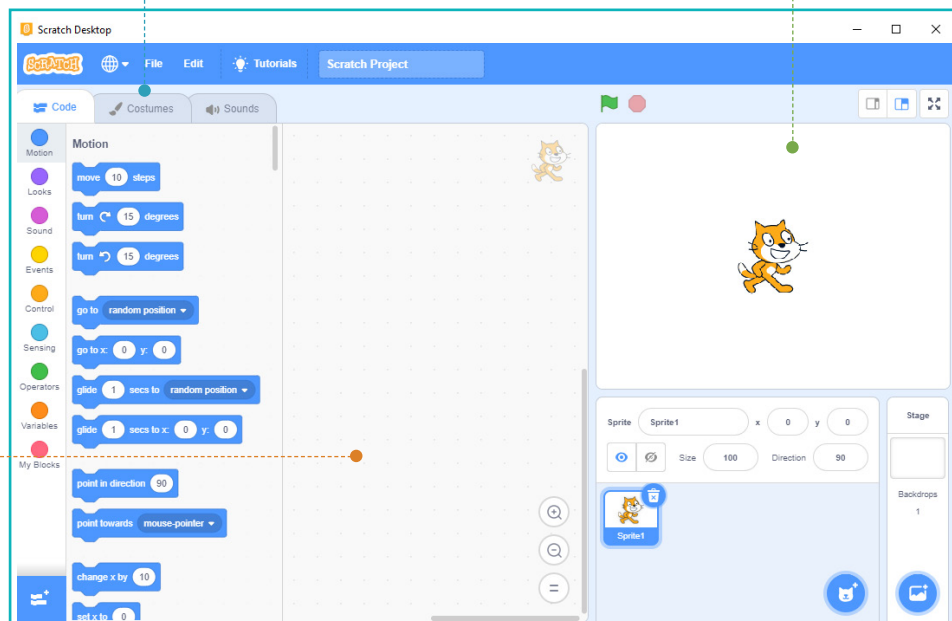


- **Code:** A code is a set of instructions which the computer follows to complete a task. Scratch uses a block-based code.
- **Stage:** The Stage is the display screen where we can see the sprites move.

Costumes: Edit sprite here

Stage: Animation runs here

Code area: Build code here



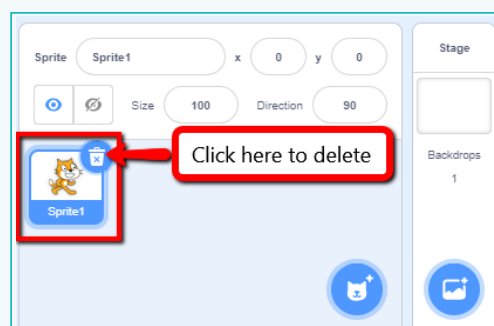
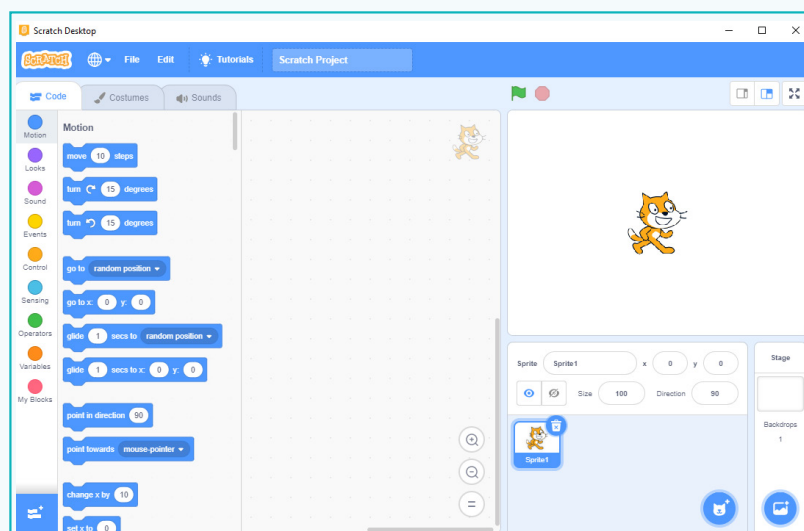
The animation we will create is called “Trip to the Moon”.

Neil Armstrong is the first man to step on the Moon. He reached the Moon on 20th July 1969. As he reached, he recorded a message for people on Earth. “Hello, I’m Neil. I’m talking to you from the moon. I have seen many things in my time here. But now, I’m coming back to Earth. See you soon!” Neil goes back into the rocket. The rocket ignites and returns to Earth.

Activity 1

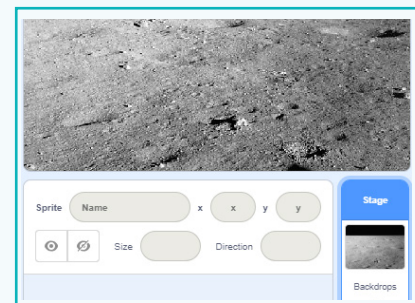
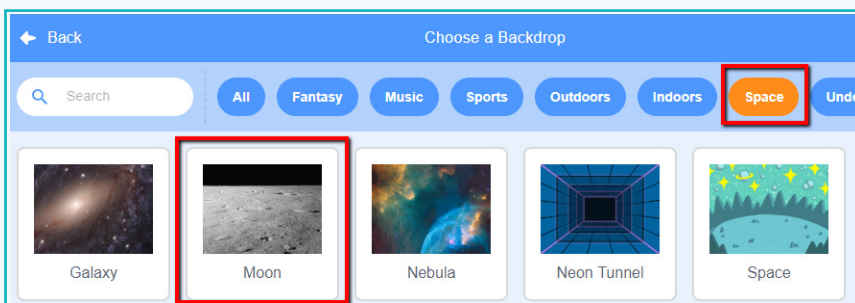
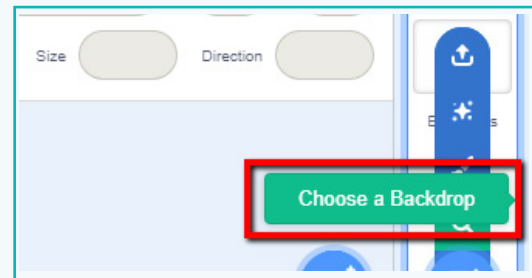
In our animation, Neil Armstrong is on the moon. We need to create a moon backdrop and a sprite for Neil Armstrong.

- 1 Open Scratch.
- 2 Delete the default cat sprite. We do not need it for this animation.

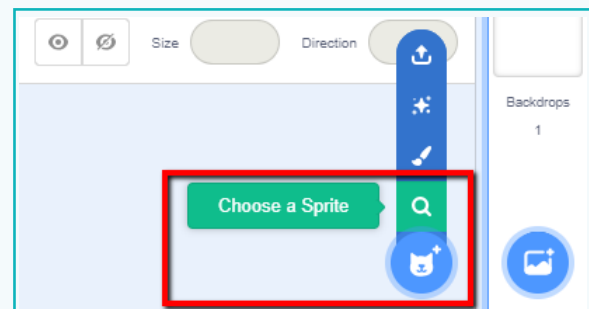


Activity 1

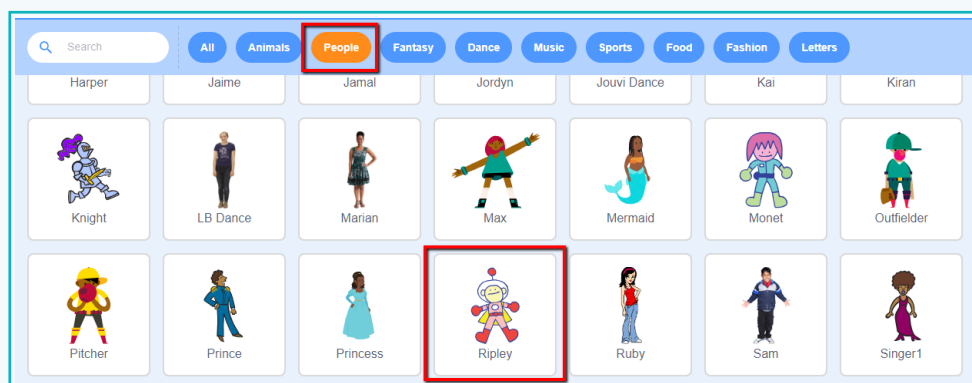
- 3 Open the backdrop library and choose the 'Moon' backdrop.



- 4 Neil Armstrong was an astronaut. We need a sprite that looks like an astronaut in a space suit.

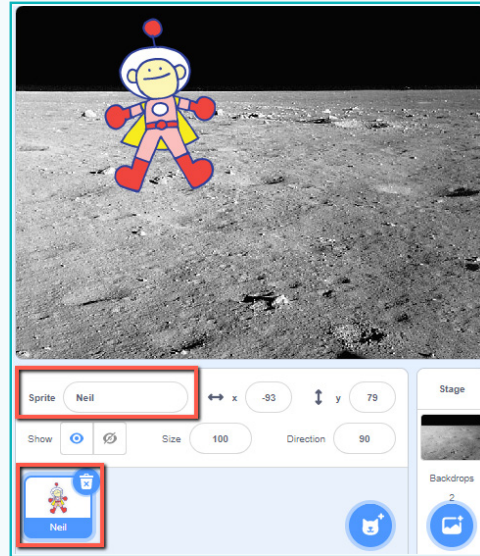


- 5 The 'Ripley' sprite looks similar to an astronaut. We will use that one.

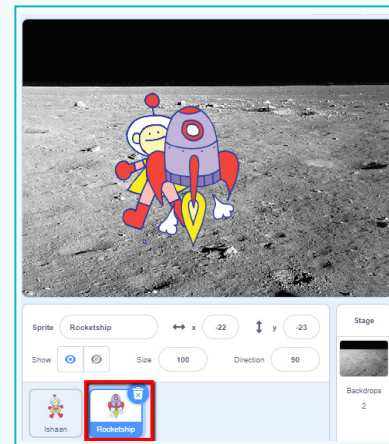
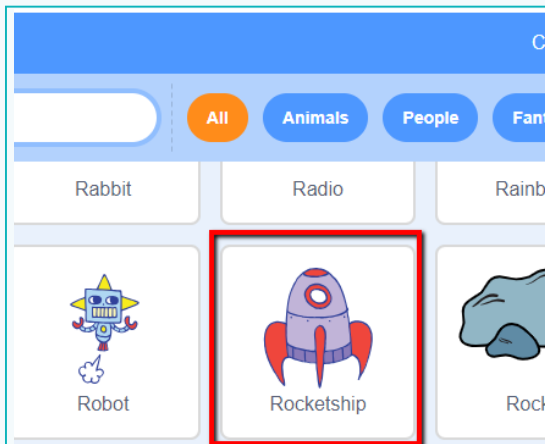


Activity 1

- 6 Change the sprite's name from 'Ripley' to 'Neil'.



- 7 Neil Armstrong returns to Earth in a rocket. For this, we will use the 'Rocketship' sprite.



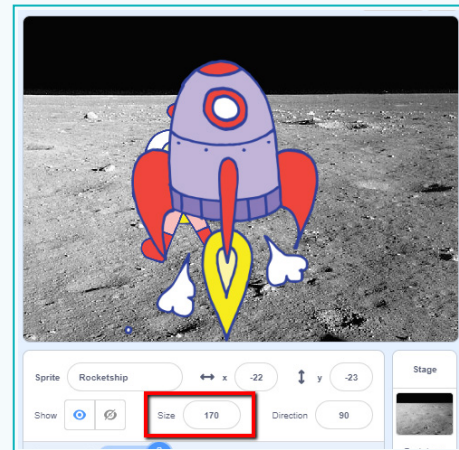
Right now, both sprites are the same size. To make it realistic, we need to increase the size of the rocket ship.

Bytes

A space suit is a garment wore in space. It helps astronauts breathe and maintain their body temperature.

Activity 1

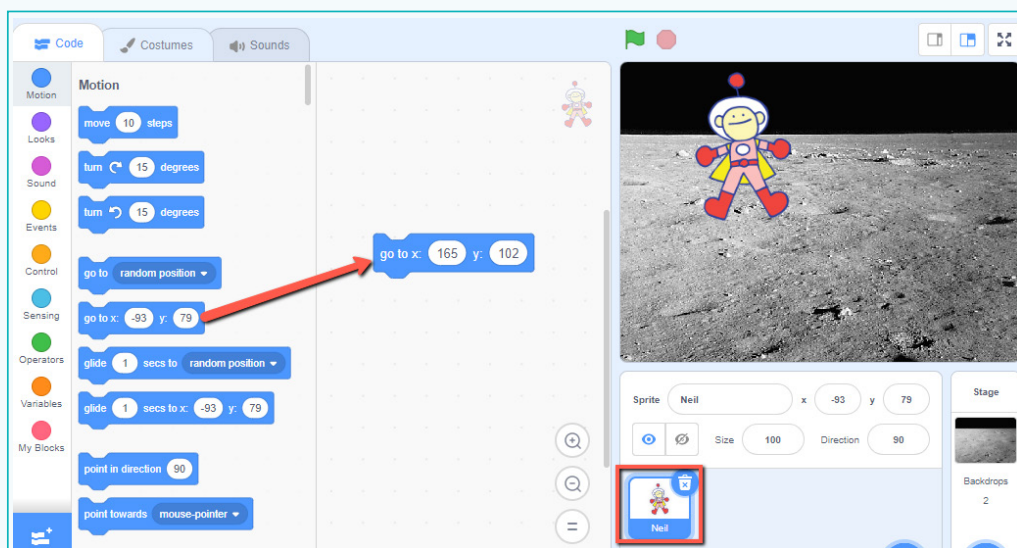
- 8 Rockets are much larger than humans. Change the size of the rocket ship to '170'.



This will be the scene for our animation. To animate it, we need to add code. Neil Armstrong has to move towards the rocket ship. To start, we need to place him away from it. We will drag Neil Armstrong's sprite towards the right. x=165, y=102 seems like a good position.

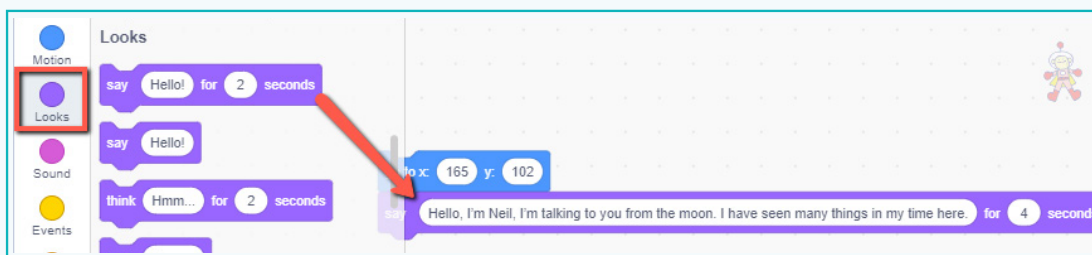
- 9 Select the Neil Armstrong's sprite and click on the 'Motion' block.
- 10 Drag the 'go to x: 165, y: 102'.

Here the values of x, y decide the position of sprite on the stage. When we move a sprite on the stage, the values of x, y change accordingly.

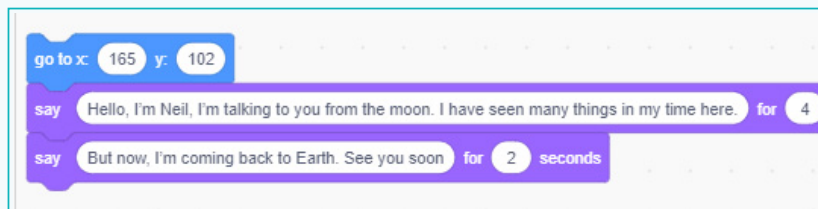


Activity 1

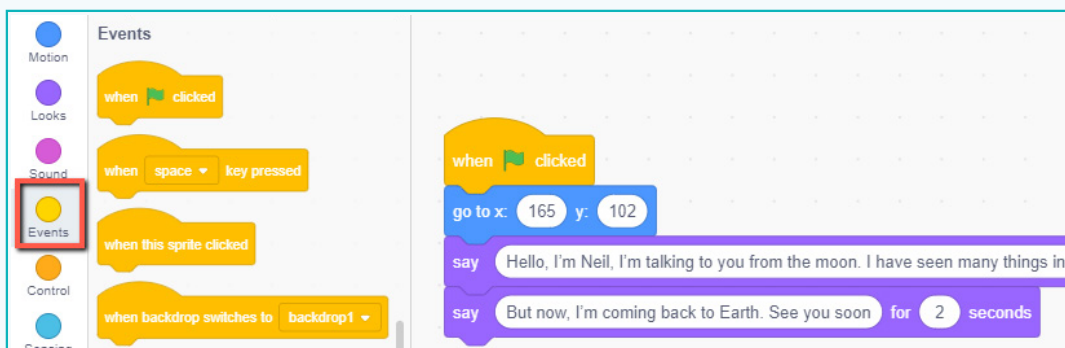
- 11 We will add the 'Say' block from the 'Looks' option for the Neil Armstrong's message to the people on Earth.
- 12 Write: "Hello I'm Neil, I'm talking to you from the Moon. I have seen many things in my time here."
- 13 The dialogue must remain on screen for a few seconds. This way, everyone will be able to read it. Set the 'time' as '4 seconds'.



- 14 Repeat the same for the second dialogue "But now, I'm coming back to Earth. See you soon".



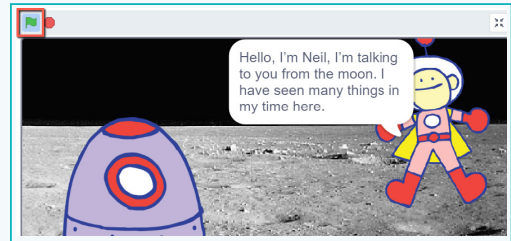
- 15 The animation should start when we click and stop when we click .
- 16 The Flag button runs the code and the animation starts. To do this, add the 'when clicked' block at the top of the code.



Activity 1

- 17 To watch the animation, click on the flag.

Here, Neil Armstrong is at the position we have fixed with his dialogues displayed.



Exercise

Tick mark ☒ the correct answer

- 1 Which block do you use to move a sprite to a specific position?

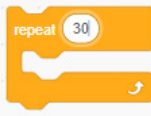
a. ☐



b. ☐



c. ☐



d. ☐

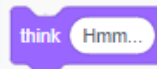


- 2 Which of the following is used to run the code?

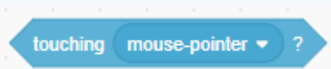
a. ☐



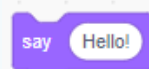
b. ☐



c. ☐



d. ☐



Fill in the blanks

- 3 To delete a sprite, _____ click on the sprite and choose delete.

- 4 is a kind of _____ block.

Activity 1

Now, let's add a code for the rocket ship.

- 18 Select the 'Rocketship' sprite.

Right now, the Rocketship sprite is fired for launch. When the animation begins, it should be stationary without any fire.

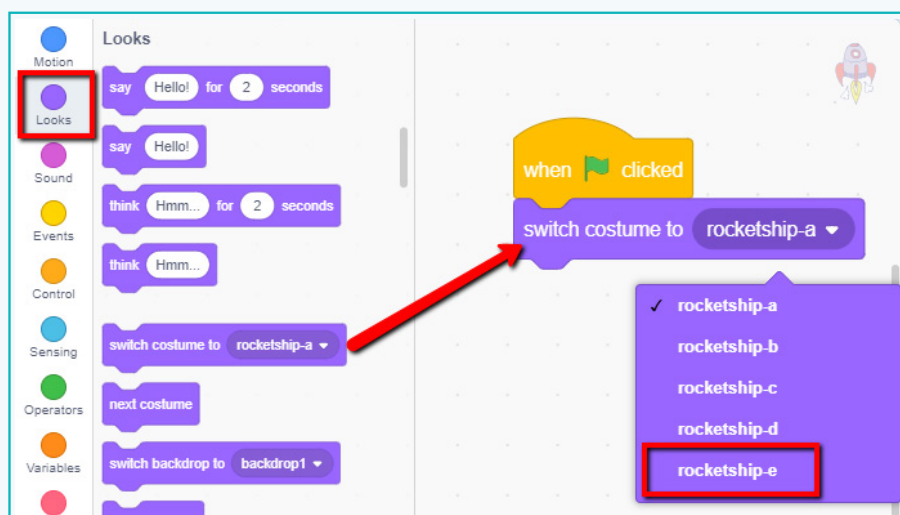
Switch costume will change the look of the sprite as per the scene. Initially we want the rocket ship without fire. For this, we need to change the costume of the Rocketship sprite.

- 19 Select the 'Rocketship' sprite.

- 20 Add 'when flag clicked' and 'switch costume to Rocketship-e'.

Note: Costume

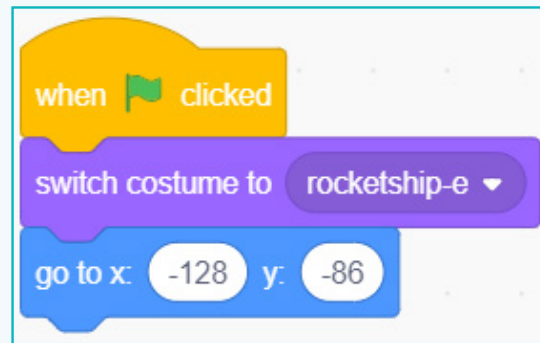
To give a sprite a different expression or position, we change its "costume". Costumes are versions of a sprite in different positions and looks.



- 21 We need to fix a position for the rocket ship. When we run the code, it should start from the fixed position.

Activity 1

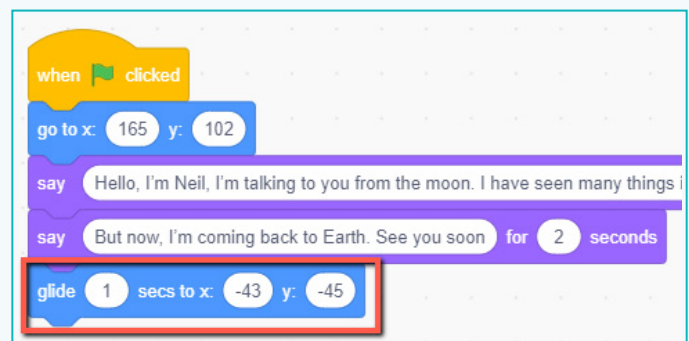
- 22 Add “go to x= -128, y= -86”.



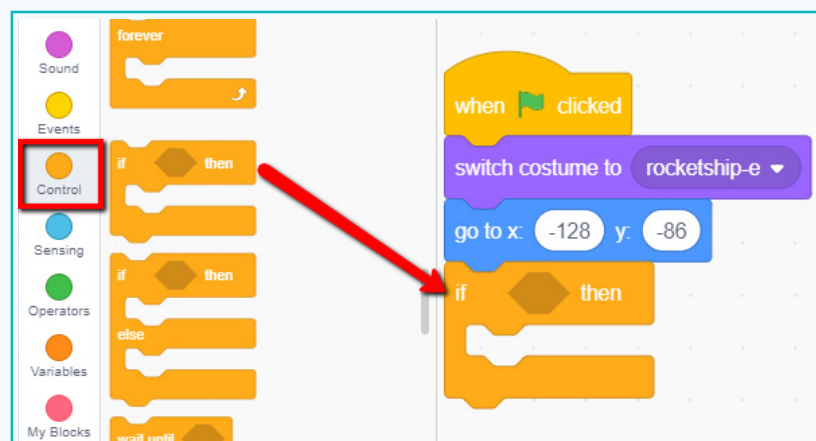
Next in our story, Neil Armstrong goes inside the rocket ship. To move Neil Armstrong from his initial position to the rocket ship, add the 'glide' block from the 'move' option.

- 23 Add 'glide' to 'x=-43, y=-45'.

- 24 We chose 'x= -43, y= -45' because it is close to the rockets current position and only 1 second is required to show the movement.

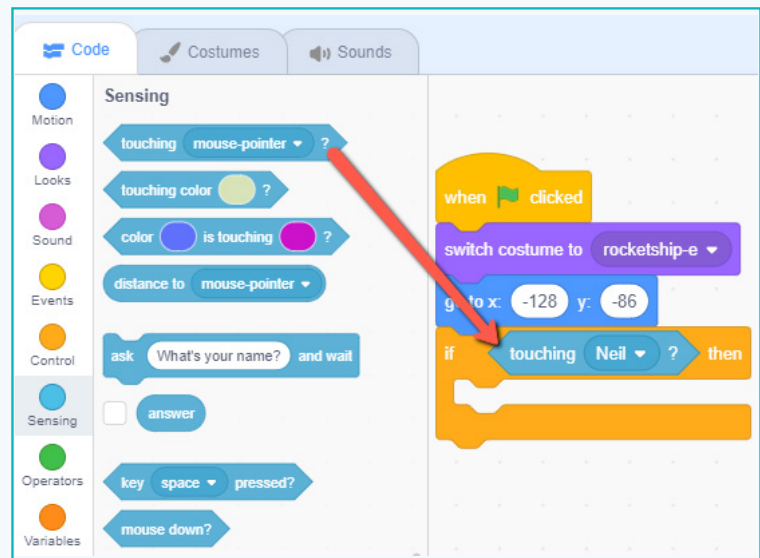


The rocket ship must fly only if Neil Armstrong touches it. For this, we will add the 'if' block in the code. The 'if' block is used to make decisions.



Activity 1

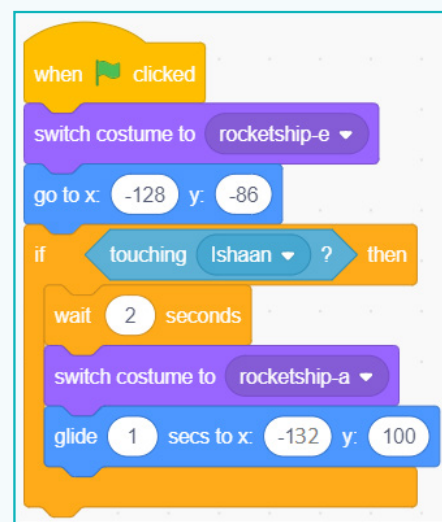
- 25 Add the condition 'if touching Neil then'.



- 26 After a 3 second wait, we will show fire under the rocket ship. Add a block to 'switch costume to Rocketship-a'.

The Rocketship must take off from the moon into the sky. To show its upward movement, we will add glide.

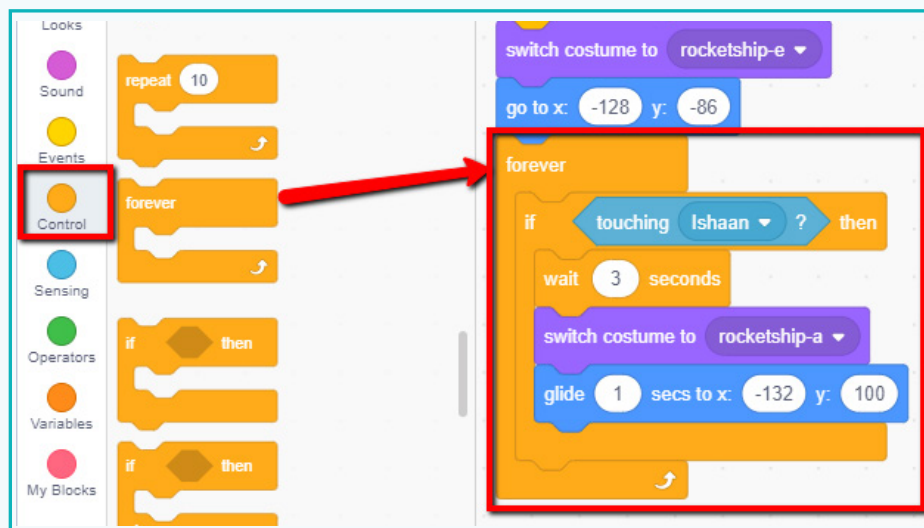
- 27 For the rocket ship to glide in an upward direction, use 'go to x= -132, y= 100'.



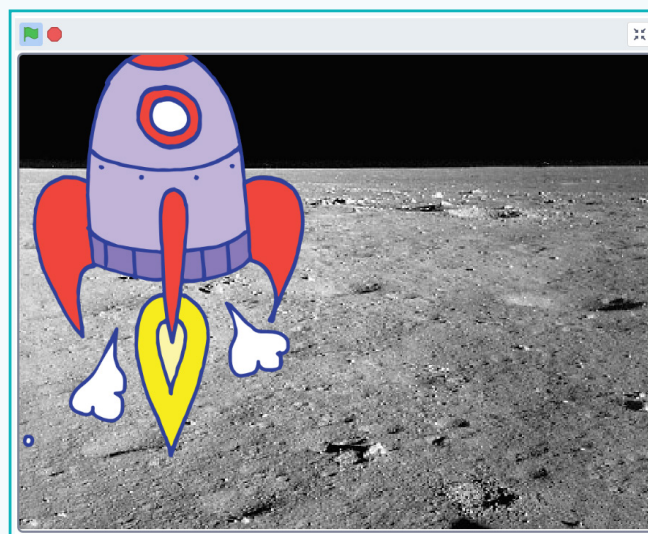
- 28 We want all the blocks inside the 'if' block to run forever.
- 29 For this, we will add the 'Forever' block above the 'if' block.

Activity 1

30 When we run the animation, the blocks held inside this will loop forever with the 'forever' block.



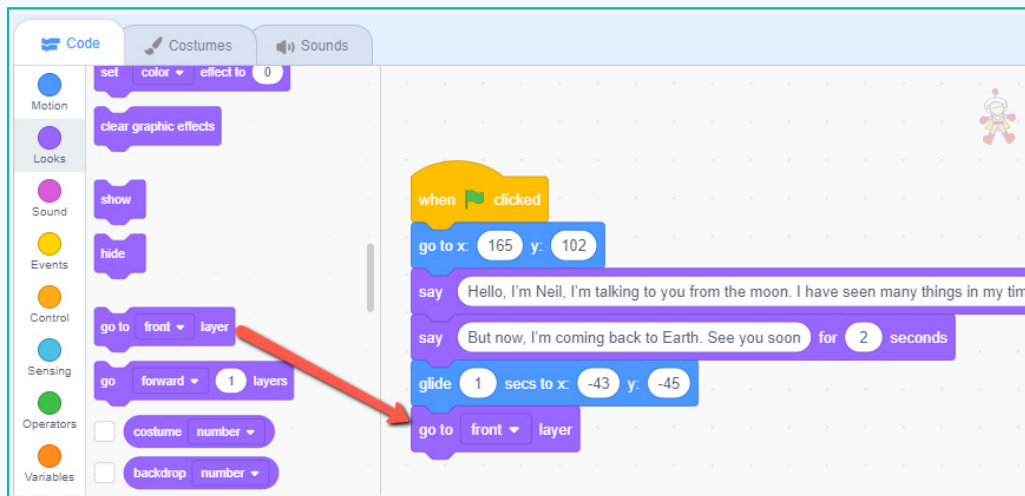
Run the animation and observe the code.



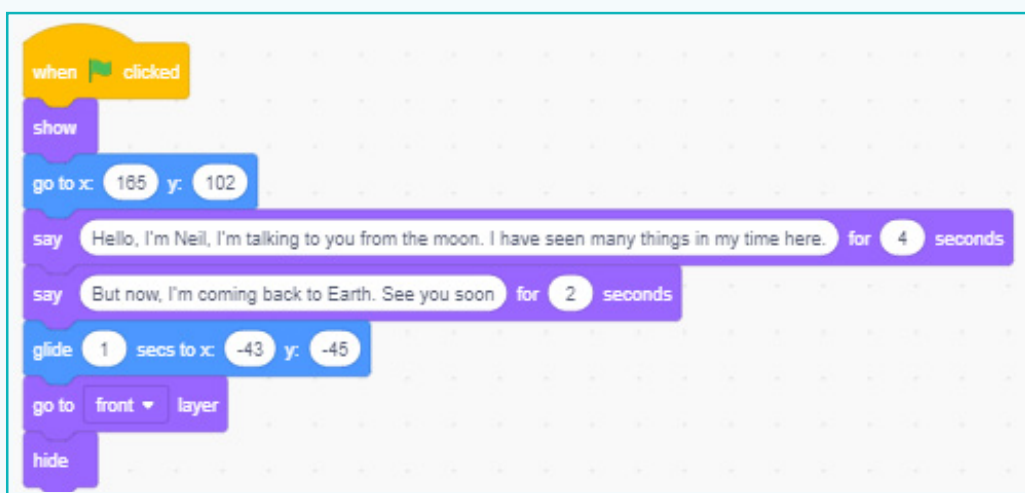
We can see that Neil Armstrong's sprite is still visible even when the rocket is flying. To correct this, we need to hide Neil Armstrong before the rocket ship flies.

Activity 1

- 31 To show that Neil Armstrong is inside the rocket ship, add the 'go to back layer' block.

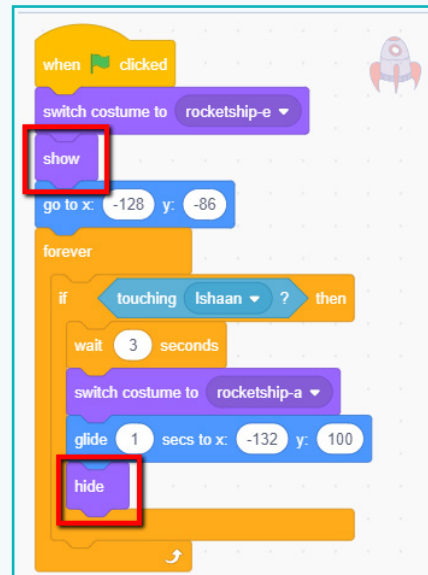


- 32 Add the 'Show' block, so when the animation starts, the sprite is visible.
- 33 Add the 'Hide' block, so when the rocket ship flies, Neil Armstrong is not visible.
- 34 Add the blocks for Neil Armstrong as shown below.

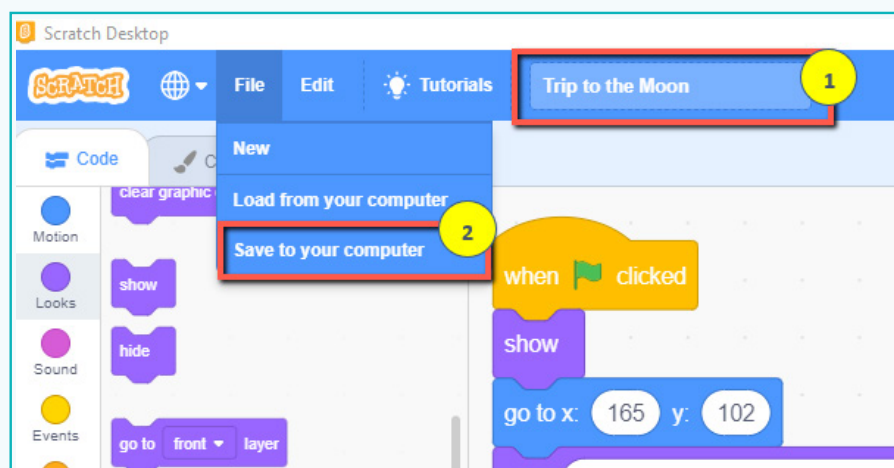


Activity 1

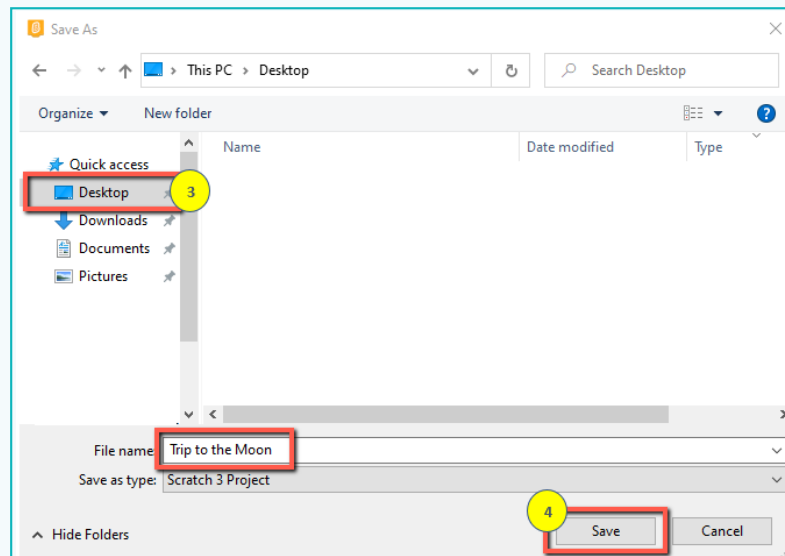
- 35 The rocket ship must also disappear once it launches and flies. Add the 'Show and 'Hide' blocks for it too.'



- 36 Run the animation to test it out.
- 37 Save the project and name it "Trip to the Moon".

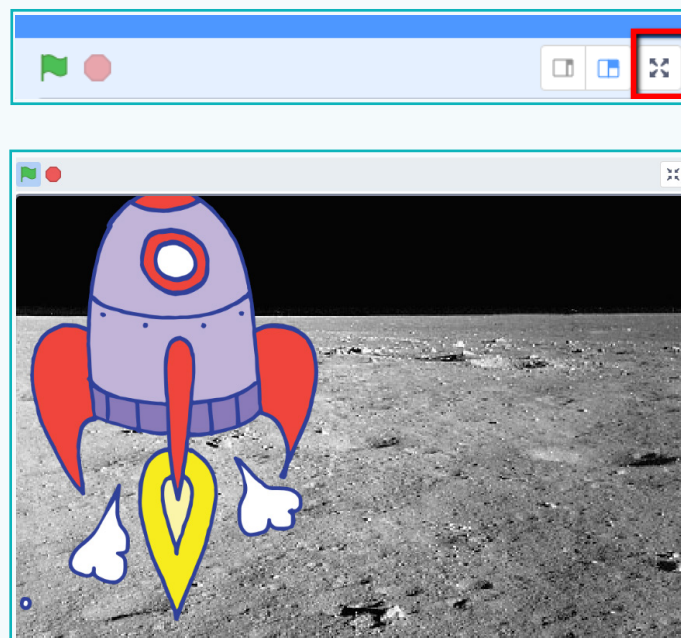


Activity 1



Run the animation and test it.

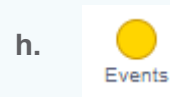
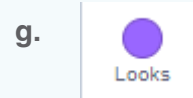
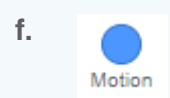
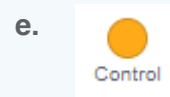
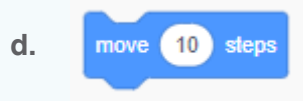
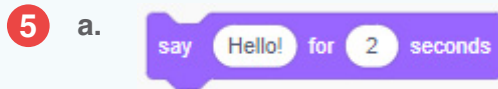
Full screen control: Full screen control is used to switch the player view to a full screen size player view.



Our “Trip to the moon” animation is complete! Use your imagination to create more stories with the “if” and “forever” blocks. Add movements and dialogues to the sprites to make the story interesting.

Exercise

5 Match the pairs



...and this is the story of Neil Armstrong, on the moon! I hope you all enjoyed it.



Tech Flash

- **Animation** : Animation is a way to add movement to drawings, models, and non-living objects. This makes them appear to move as if they are alive.
- **Duplicate** : To duplicate means to make an exact copy of something.
- **Costume** : To give a sprite a different look or position, we can change its "costume". A costume is a version of a sprite in different poses and expression.
- **Code** : A code is a set of step wise instructions given to the sprite to complete a particular task.
- **Motion block** : The Motion block contains instructions to make the sprite move, glide, take a number of steps, move in a specific direction, etc.
- **Looks block** : The Looks block contains instructions to change the appearance of the sprite. E.g.: switch costume, change colour, say, etc.
- **Full screen control** : Full screen control is used to switch the player view to a full screen size player view.